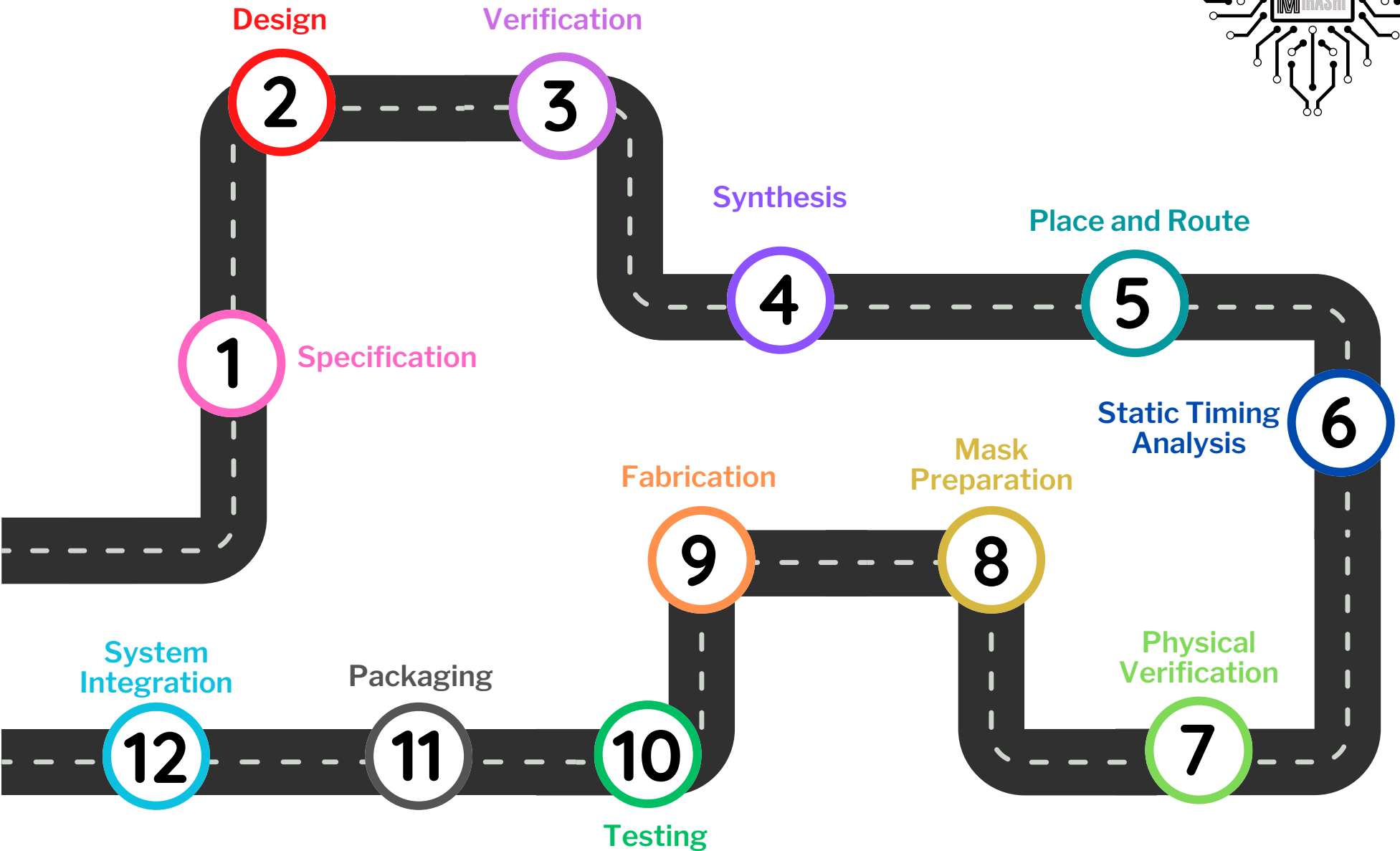
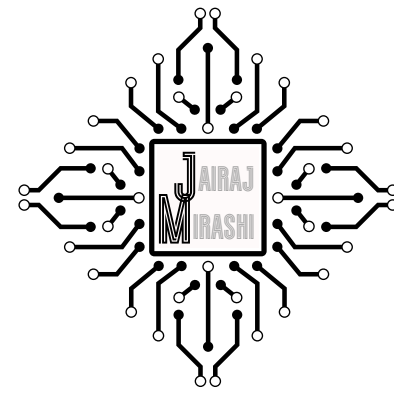


STEP BY STEP

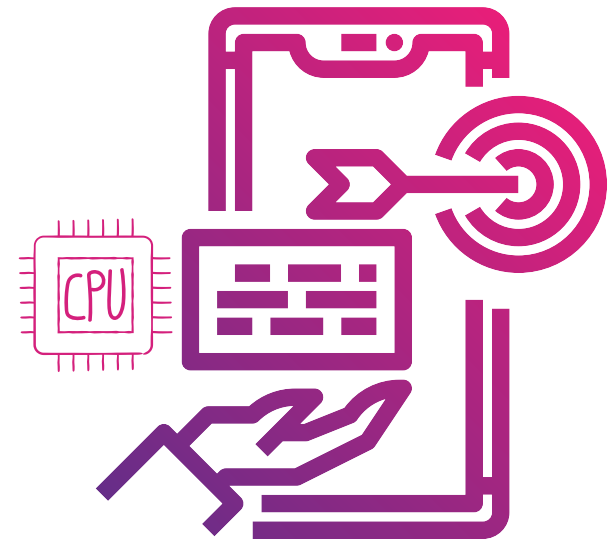


1

SPECIFICATION

IN THE SPECIFICATION STAGE, THE REQUIREMENTS FOR THE ASIC ARE DEFINED.

THIS INCLUDES FUNCTIONALITY, PERFORMANCE, POWER CONSUMPTION, AND OTHER KEY PARAMETERS. THE SYSTEM ARCHITECT AND THE CUSTOMER WORK ON DEFINING THE SPECIFICATION.



2

DESIGN

IN THE DESIGN STAGE, THE DIGITAL OR ANALOG CIRCUIT DESIGN IS CREATED USING COMPUTER-AIDED DESIGN (CAD) TOOLS. THE DIGITAL DESIGN ENGINEER WORKS ON THE LOGICAL DESIGN OF THE CIRCUITS, WHILE THE ANALOG DESIGN ENGINEER WORKS ON THE DESIGN OF THE ANALOG AND MIXED-SIGNAL CIRCUITS. THE DESIGN PROCESS INVOLVES VERIFYING THE FUNCTIONALITY, TIMING, POWER CONSUMPTION, AND OTHER PERFORMANCE METRICS OF THE CIRCUIT USING SIMULATION TOOLS.



3

VERIFICATION

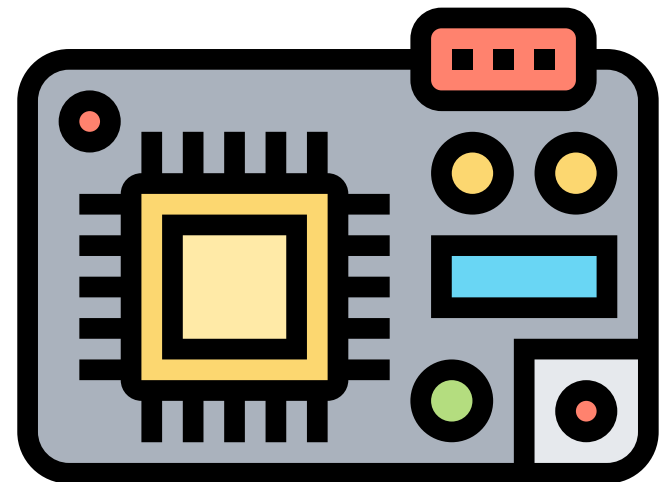
IN THE VERIFICATION STAGE, ENGINEERS USE SIMULATION AND OTHER TOOLS TO ENSURE THAT THE DESIGN WORKS CORRECTLY AND MEETS THE SPECIFICATIONS. VERIFICATION ENGINEERS WORK ON TESTING THE DESIGN AT DIFFERENT LEVELS OF ABSTRACTION, INCLUDING FUNCTIONAL, TIMING, AND POWER. THEY USE VARIOUS TECHNIQUES SUCH AS SIMULATION, FORMAL VERIFICATION, AND EMULATION TO ENSURE THAT THE DESIGN IS CORRECT AND ROBUST.



4

SYNTHESIS

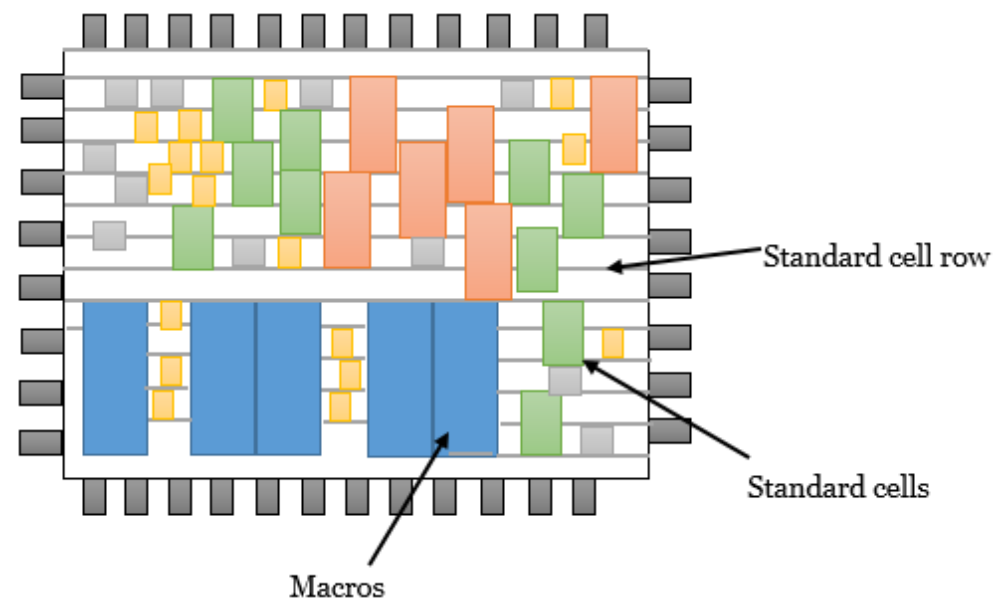
IN THE SYNTHESIS STAGE, THE DESIGN IS TRANSLATED INTO A NETLIST, WHICH IS A LIST OF LOGIC GATES AND THEIR CONNECTIONS. THIS NETLIST IS OPTIMIZED FOR AREA, TIMING, AND POWER USING SYNTHESIS TOOLS. THE SYNTHESIS ENGINEER WORKS ON OPTIMIZING THE DESIGN FOR PERFORMANCE AND AREA.



5

PLACE AND ROUTE

IN THE PLACE AND ROUTE STAGE, THE NETLIST IS MAPPED ONTO A PHYSICAL LAYOUT OF THE ASIC. THIS INVOLVES PLACING THE CELLS AND ROUTING THE WIRES BETWEEN THEM. THE PLACE AND ROUTE ENGINEER WORKS ON OPTIMIZING THE PLACEMENT AND ROUTING FOR TIMING, AREA, AND POWER.



6 STATIC TIMING ANALYSIS

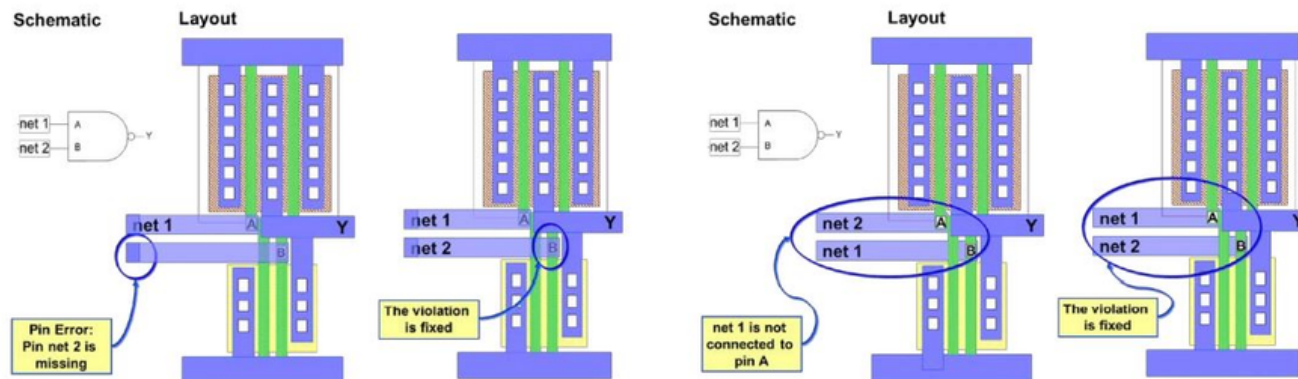
IN THE STATIC TIMING ANALYSIS STAGE, THE TIMING OF THE DESIGN IS ANALYZED TO ENSURE THAT IT MEETS THE TIMING REQUIREMENTS. THE STATIC TIMING ANALYSIS ENGINEER WORKS ON ANALYZING THE TIMING OF THE DESIGN AND ENSURING THAT IT MEETS THE TIMING CONSTRAINTS.



7

PHYSICAL VERIFICATION

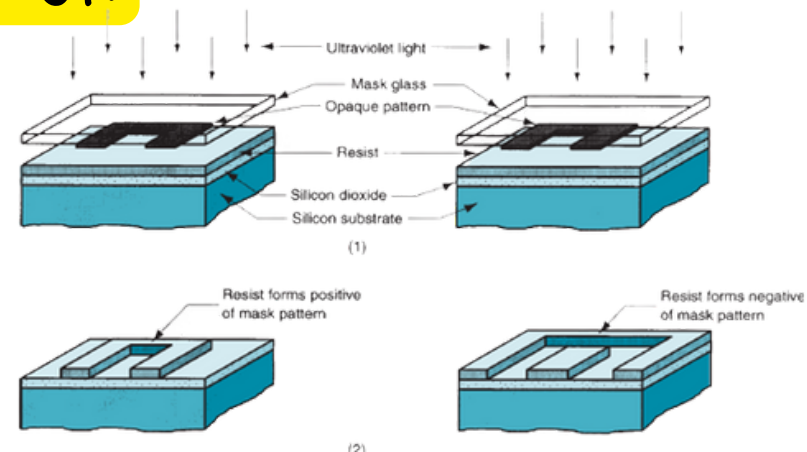
IN THE PHYSICAL VERIFICATION STAGE, THE PHYSICAL LAYOUT OF THE ASIC IS VERIFIED TO ENSURE THAT IT IS CORRECT AND MEETS THE DESIGN RULES. THIS INVOLVES CHECKING FOR DESIGN RULE VIOLATIONS, SUCH AS MINIMUM SPACING AND MINIMUM WIDTH VIOLATIONS. THE PHYSICAL VERIFICATION ENGINEER WORKS ON ENSURING THAT THE LAYOUT IS CORRECT AND CAN BE MANUFACTURED.



8

MASK PREPARATION

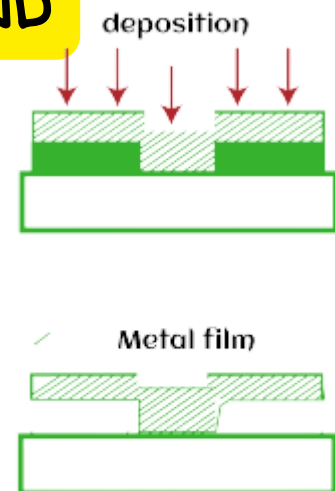
IN THE MASK PREPARATION STAGE, THE PHYSICAL LAYOUT IS USED TO CREATE THE MASK SET, WHICH IS A SET OF PATTERNS THAT WILL BE USED TO CREATE THE ACTUAL CIRCUIT ON THE SILICON WAFER. THE MASK PREPARATION ENGINEER WORKS ON CREATING THE CORRECT PATTERN FOR EACH LAYER OF THE CHIP. THIS INVOLVES CONVERTING THE LAYOUT INTO A SET OF PHOTOLITHOGRAPHY MASKS, WHICH ARE USED TO PATTERN THE SILICON WAFER. THE MASK PREPARATION ENGINEER NEEDS TO ENSURE THAT THE MASKS ARE CORRECTLY ALIGNED AND THAT THE PATTERNS ARE OPTIMIZED FOR THE PROCESS TECHNOLOGY.



9

FABRICATION

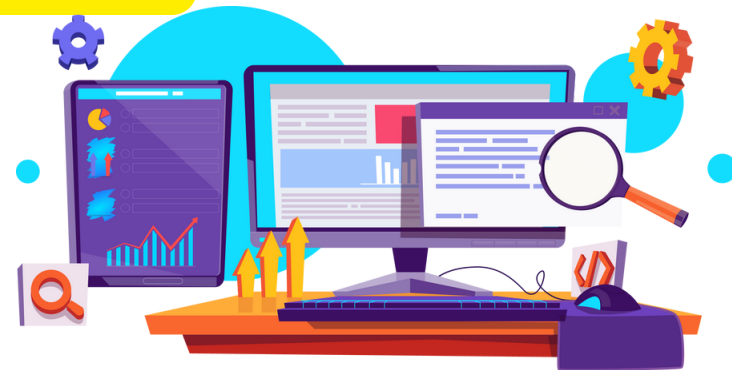
IN THE FABRICATION STAGE, THE MASK SET IS USED TO CREATE THE ASIC ON THE SILICON WAFER. THE FABRICATION PROCESS INVOLVES SEVERAL STEPS, SUCH AS PHOTOLITHOGRAPHY, DEPOSITION, ETCHING, AND IMPLANTATION. THE FABRICATION ENGINEER WORKS ON THE PROCESS OF CREATING THE DIFFERENT LAYERS OF THE ASIC USING THESE TECHNIQUES. THE FABRICATION PROCESS IS HIGHLY AUTOMATED AND REQUIRES A CLEAN-ROOM ENVIRONMENT TO ENSURE THE QUALITY AND RELIABILITY OF THE ASIC.



10

TESTING

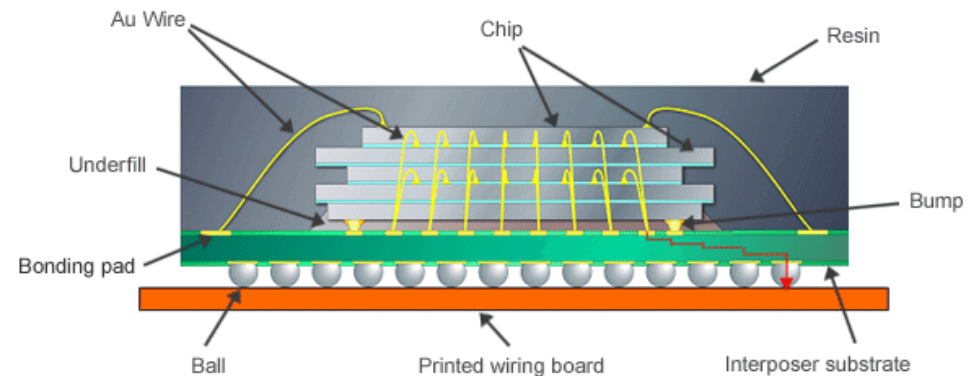
IN THE TESTING STAGE, THE ASIC IS TESTED TO ENSURE THAT IT WORKS CORRECTLY AND MEETS THE SPECIFICATIONS. TESTING ENGINEERS WORK ON DEVELOPING TEST PLANS AND TEST PATTERNS TO VERIFY THE FUNCTIONALITY AND PERFORMANCE OF THE ASIC. THEY USE VARIOUS TECHNIQUES SUCH AS AUTOMATIC TEST EQUIPMENT (ATE), SCAN CHAINS, AND BUILT-IN SELF-TEST (BIST) TO TEST THE ASIC. THE TESTING PROCESS IS CRITICAL TO ENSURING THE QUALITY AND RELIABILITY OF THE ASIC.



11

PACKAGING

IN THE PACKAGING STAGE, THE ASIC IS ENCAPSULATED IN A PACKAGE THAT PROVIDES ELECTRICAL CONNECTIONS TO THE OUTSIDE WORLD AND PROTECTS THE ASIC FROM ENVIRONMENTAL FACTORS SUCH AS MOISTURE, TEMPERATURE, AND PHYSICAL DAMAGE. THE PACKAGING ENGINEER WORKS ON SELECTING THE APPROPRIATE PACKAGE FOR THE ASIC AND DESIGNING THE LAYOUT OF THE PACKAGE. THE PACKAGING PROCESS ALSO INVOLVES TESTING THE ASIC AFTER IT IS PACKAGED TO ENSURE THAT IT STILL WORKS CORRECTLY.



12 SYSTEM INTEGRATION

IN THE SYSTEM INTEGRATION STAGE, THE ASIC IS INTEGRATED INTO THE LARGER SYSTEM IN WHICH IT WILL BE USED. THIS INVOLVES DESIGNING THE BOARD OR SYSTEM THAT THE ASIC WILL BE MOUNTED ON, AS WELL AS WRITING THE SOFTWARE THAT WILL RUN ON THE SYSTEM. SYSTEM INTEGRATION ENGINEERS WORK ON INTEGRATING THE ASIC INTO THE LARGER SYSTEM, TESTING ITS FUNCTIONALITY, AND OPTIMIZING ITS PERFORMANCE IN THE SYSTEM.

